

MASTER'S COMPREHENSIVE EXAMINATION

Part I Theory (2 hours) December 3rd, 2016

Instructions: Attempt any three out of the following four questions on both pages.

Mention which three you would like to have graded. Open book and open notes.

1. Suppose that $X_1, X_2, \dots, X_n$ are n independent random variables such that X_i is distributed by a Poisson distribution with mean $\mu_i = \lambda_i \theta$, where the λ_i 's are given constants.

- (I) Please calculate both the maximum likelihood estimator (MLE) of θ and the least squares (LS) estimator of θ .
 - (II) Is the maximum likelihood estimator of θ unbiased? If not so, please calculate the bias.
 - (III) Calculate also the bias of the least square estimator of θ . Compare the biases of these two estimators. Is the bias a good criterion for comparing two estimators or should we look at some other criteria that also involves the variance?
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2. Let $X_i, i = 1$ to n , be i.i.d. samples from $N(\mu, \sigma^2)$. Let $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. Prove that $\bar{X}$ and $X_i - \bar{X}$ are uncorrelated for any i .

3. Suppose we have 10 coins such that if the i -th coin is flipped, heads will appear with probability $i/10, i = 1, 2, \dots, 10$.

- (i) If all 10 coins are flipped, what is the expected number of heads?
 - (ii) When one of the coins is randomly selected and flipped, it shows head. What is the conditional probability that it was the fifth coin, i.e., $P\{5\text{th}|\text{heads}\}$?
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4. Suppose that Y has a log-normal distribution with density $f(y) = \frac{1}{y\sqrt{2\pi\sigma^2}} e^{-\frac{(\log(y)-\mu)^2}{2\sigma^2}}, y > 0$

- (i) Show that $Z = \log(Y)$ has a $Normal(\mu, \sigma^2)$ distribution.
- (ii) Calculate the expected value and variance of Y .
- (iii) Show that if $Y_1, Y_2, \dots, Y_n$ are independent and with the same distribution as Y then $\prod_{i=1}^n Y_i$ has also a log-normal distribution.